Church Analytics Dashboard — Cloud Migration Plan

Context

DUMC (a church) is migrating its existing analytics dashboards (Power BI + NCG vendor-built) to Alibaba Cloud. The primary data source is **PCO (Planning Center Online)**, synced daily via API into a PostgreSQL/RDS database. A secondary source (**Chinese Power BI**) provides supplementary CN congregation data. Scale is medium (1-50GB) with daily/weekly refresh. The BI tool will be either Power BI or a new vendor (TBD). Everything must stay within Alibaba Cloud. Budget-conscious — no AnalyticDB needed.

Data sources:

- **PCO API** → `pco_*` tables (users, groups, events, attendance, check-ins, decisions, discipleship, serving)
- **Chinese Power BI** → supplementary CN-specific CG and serving data
- **User Requirements** → 7 modules defined in "Imperative 10: Analytic - User Requirement" document

Recommended Architecture: Lightweight 3-Layer on RDS + Cron Job

```
┌──────────────────────────────────────────────────────────┐
│                    PRESENTATION LAYER                    │
│              Power BI / QuickBI / New Vendor             │
└──────────────────────────┬───────────────────────────────┘
                           │ reads from
┌──────────────────────────▼───────────────────────────────┐
│                SEMANTIC LAYER                            │
│           Separate "sem" schema in RDS                   │
│          Star schema: fact tables + dimension tables     │
│          Pre-aggregated, query-optimized                 │
└──────────────────────────┬───────────────────────────────┘
                           │ ETL (daily)
┌──────────────────────────▼───────────────────────────────┐
│                   TRANSFORM LAYER                        │
│              Cron Job + SQL Scripts (ETL)                │
│          Clean, join, dedupe, business logic             │
└──────────────────────────┬───────────────────────────────┘
                           │ reads from
┌──────────────────────────▼───────────────────────────────┐
│                 RAW / LANDING LAYER                      │
│              RDS / ApsaraDB (source data)                │
│          Data as-is from migration, no transforms        │
└──────────────────────────────────────────────────────────┘
```

What is a Star Schema?

A star schema is a **data model** optimized for analytics and reporting. It organizes tables into two types — **fact tables** (measurements) and **dimension tables** (descriptors) — with the fact table in the center and dimensions radiating outward like a star.

```
                      ┌──────────────┐
                      │   dim_date   │
                      │    (WHEN)    │
                      └──────┬───────┘
                             │
┌──────────────┐      ┌──────▼───────────┐      ┌────────────────────┐
│ dim_member   │──────│   FACT TABLES    │──────│dim_congregation    │
│    (WHO)     │      │                  │      │ (WHICH CONGR.)     │
└──────────────┘      └──────────────────┘      └────────────────────┘
                             │
                      ┌──────▼───────────┐
                      │ dim_cell_group   │
                      │(WHICH CG/ZONE)   │
                      └──────────────────┘
```

Summary: 4 Dimensions + 6 Facts

Dimensions (descriptors — filter & group by):

Table	Source	Key Columns	Design Notes
`dim_date`	generated	year, quarter, month, week_of_year, day_of_week, weekend_flag	YYYYMMDD int key, ~3,650 rows for 10 years
`dim_congregation`	seed data (8 rows)	congregation_code, congregation_label	EN/BM/CN/MM/NP/TM/FP/NXG — no ETL needed
`dim_member`	`pco_users`	full_name, gender, age_group, membership, user_role, is_baptized, congregation_key	member_key = user_id (no surrogate key)
`dim_cell_group`	`pco_groups` (type 137505)	cg_name, zone, district, subzone, zone_leader, leader_member_key, age_group, frequency, cg_size, status, congregation_key	Zone/district/subzone denormalized here (no separate dim_zone — avoids extra joins)

Facts (measures — count & aggregate):

Table	Grain	Key Measures	Pillar	PCO Source
`fact_cg_attendance`	member × CG event	attended (0/1), user_role (member/visitor/leader)	Cell Group	`pco_attendance` + `pco_events`
`fact_cg_submission`	CG × event	submitted (0/1), cancelled (0/1)	Cell Group	`pco_events` (event_att_committed_at)
`fact_decisions`	decision form	form_id (112029/117917), form_title, congregation_key	Evangelism	`pco_new_decisions`
`fact_discipleship`	participant	status (In progress/Completed), started_date_key, completed_date_key	Evangelism, Training	`pco_discipleship_completion`
`fact_celebration_attendance`	check-in	kind (Guest/Regular/Volunteer), service_name, congregation_key	Celebration	`pco_checkin_attendance`
`fact_serving`	volunteer × date	team_name, ministry, position	Serving	`pco_serving_attendance`

Key design choices:

- **No `dim_zone`** — zone/district/subzone are denormalized on `dim_cell_group` (avoids extra join for every CG query)
- **`dim_congregation` is a tiny seed table** (8 rows) — no ETL, just reference data
- **`member_key` = `user_id`** directly from PCO — no surrogate key generation
- All fact tables use `date_key` (INT YYYYMMDD) for consistent time slicing
- SQL definition: `02-semantic-layer/01-create-sem-schema.sql`

Why Not Just Query Raw Tables Directly?

Raw tables	Star schema
Normalized (many JOINs needed)	Pre-joined, fewer JOINs
No computed fields	Age groups, engagement tiers pre-calculated
Codes like `english`	Friendly labels like "English Congregation"
No date dimension	Easy to slice by week, month, quarter, year
Optimized for data entry	Optimized for dashboards

Examples

Weekly CG Attendance by Zone (Cell Group pillar):

```sql
SELECT cg.zone, d.year, d.week_of_year,
       SUM(f.attended) AS total_attended,
       COUNT(*) AS total_records
FROM sem.fact_cg_attendance f
JOIN sem.dim_cell_group cg ON f.cg_key = cg.cg_key
JOIN sem.dim_date d ON f.date_key = d.date_key
WHERE f.attended = 1
GROUP BY cg.zone, d.year, d.week_of_year;
```

CG Submission Rate (Cell Group pillar):

```
SELECT dc.congregation_label, d.year, d.week_of_year,
       SUM(f.submitted) AS submitted_count,
       COUNT(*) AS total_events,
       ROUND(SUM(f.submitted) * 100.0 / COUNT(*), 1) AS submission_rate_pct
FROM sem.fact_cg_submission f
JOIN sem.dim_cell_group cg ON f.cg_key = cg.cg_key
JOIN sem.dim_congregation dc ON cg.congregation_key = dc.congregation_key
JOIN sem.dim_date d ON f.date_key = d.date_key
WHERE f.cancelled = 0
GROUP BY dc.congregation_label, d.year, d.week_of_year;
```

Decisions by Congregation (Evangelism pillar):

```
SELECT dc.congregation_label, d.year, d.month,
       COUNT(*) AS total_decisions
FROM sem.fact_decisions f
JOIN sem.dim_congregation dc ON f.congregation_key = dc.congregation_key
JOIN sem.dim_date d ON f.date_key = d.date_key
GROUP BY dc.congregation_label, d.year, d.month;
```

Celebration Attendance by Kind (Celebration pillar):

```
SELECT dc.congregation_label, d.year, d.week_of_year,
       SUM(CASE WHEN f.kind = 'Regular' THEN 1 ELSE 0 END) AS regular_count,
       SUM(CASE WHEN f.kind = 'Guest' THEN 1 ELSE 0 END) AS guest_count,
       SUM(CASE WHEN f.kind = 'Volunteer' THEN 1 ELSE 0 END) AS volunteer_count,
       COUNT(*) AS total_checkins
FROM sem.fact_celebration_attendance f
JOIN sem.dim_congregation dc ON f.congregation_key = dc.congregation_key
JOIN sem.dim_date d ON f.date_key = d.date_key
GROUP BY dc.congregation_label, d.year, d.week_of_year;
```

Serving Frequency Tiers (Serving pillar):

```
SELECT
    CASE
        WHEN serve_count >= 4 THEN 'High (4+/month) - burnout risk'
        WHEN serve_count BETWEEN 1 AND 3 THEN 'Moderate (1-3/month) - consistent'
        ELSE 'Low (0/month) - capacity available'
    END AS frequency_tier,
    COUNT(*) AS member_count
FROM (
    SELECT f.member_key, COUNT(*) AS serve_count
    FROM sem.fact_serving f
    JOIN sem.dim_date d ON f.date_key = d.date_key
    WHERE d.year = 2026 AND d.month = 3
    GROUP BY f.member_key
) monthly
GROUP BY frequency_tier;
```

The dimension keys (`member_key`, `date_key`, `cg_key`, `congregation_key`) connect everything — they are the "arms" of the star. The daily ETL process keeps the star schema updated so dashboard queries stay fast and simple.

Layer Details

4. Presentation Layer — BI Tool

- **Power BI**: Can connect to RDS via MySQL connector. Works well.
- **QuickBI**: Alibaba's native tool, tightest integration, lower cost.
- **New vendor**: As long as they support MySQL protocol, they can connect.

DUMC 7 Pillars & Dashboard Metrics

The semantic layer is structured around DUMC's **7 pillars**. Each pillar has specific dashboard reports sourced from **NCG** (vendor system) and/or **Chinese Power BI**.

Pillar 1: Membership

> *From User Requirement Module 1: "In XX congregation, what is the ratio of official church members, to cell group members, and to lost sheep?"*

Dashboard Report	Source	Key Metrics (from PCO)
DUMC CG Size by Year	PCO	`pco_groups` (type_key=137505) grp_size by year
Staff Demographics	PCO	`pco_users` filtered by user_role containing 'Pastor'/'Elder' + staff list
Pastor Demographics	PCO	`pco_users` where user_role IN ('Pastor', 'Zone Pastor', etc.)
CG Leader Demographics	PCO	`pco_users` where user_role = 'Cell Group Leader'
CG Leader Age / Gender	PCO	Above + parsed user_birthdate → age_group, user_gender
Church Member % (TRAC)	PCO	user_membership='Church Member' ÷ total (Church+Cell+Visitor)
CG Member %	PCO	user_membership='Cell Member' ÷ total
Lost Sheep %	PCO	Church Members with no CG attendance in past 6 months ÷ total

Pillar 2: Cell Group

> *From User Requirement Module 2: "How consistent and healthy is member engagement and leadership accountability across our Cell Groups?"*

Dashboard Report	Source	Key Metrics (from PCO)
Weekly CG Attendance [All Congregation]	PCO	`pco_attendance` att_status='1' WHERE group type_key=137505, by week
CG Health Analysis EN/BM/CN/MM/NP/TM	PCO	Attendance %, submission %, visitor count, leader visits — per congregation
Weekly CG Attendance Analysis By Zone	PCO	Attendance by `pco_groups.grp_tag_zone`
Average CG Attendance By Zone	PCO	AVG attendance per zone over time
Weekly CG Attendance Rate By CG	PCO	Attended ÷ total CG members per group per week
CG Submission Rate	PCO	Events where event_att_committed_at IS NOT NULL ÷ total CG events
Non-submission of CG Attendance	PCO	CG events WHERE event_att_committed_at IS NULL (not submitted)
Pastor Attendance Analysis	PCO	`pco_attendance` where att_user_role='leader' AND user_role='Pastor'
Zone Leader CG Attendance Analysis	PCO	Attendance where att_user_role='visitor' AND user_role='Zone Leader'
Weekly Count of Active CG Sessions	PCO	COUNT of CG events WHERE event_canceled='0' per week
CG Scheduled %	PCO	Total CG events ÷ total weeks × number of CGs
CG Cancelled count	PCO	COUNT WHERE event_canceled='1'
Visitor Visitation	PCO	att_user_role='visitor' AND user_membership='Visitor' count per CG
CN Cell Group metrics	Chinese Power BI	CN-specific CG data (supplementary)
CN CG Attendance	Chinese Power BI	CN-specific attendance (supplementary)

Pillar 3: Evangelism

> *From User Requirement Module 3: "How many people have made the decision to accept Christ, been baptized, or finished Discipleship 101?"*

Dashboard Report	Source	Key Metrics (from PCO)
DUMC Decisions Churchwide	PCO	`pco_new_decisions` COUNT(*) — all forms
BM/CN/EN/MM/NP/TM/FP Decisions	PCO	Decisions by congregation (parsed from form context/user's group)
EN NXG CC Decisions	PCO	Decisions filtered by NXG/CC program context
Baptisms by Congregation	PCO	`pco_users` WHERE user_baptized IS NOT NULL, by congregation
D101 Completions	PCO	`pco_discipleship_completion` WHERE dc_workflow_completion='Completed'
Decision Forms: Outside (112029) vs Online (117917)	PCO	COUNT by nd_form_submitted value

Pillar 4: Serving

> *From User Requirement Module 5: "What proportion of people are actively serving, and how often are they serving?"*

Dashboard Report	Source	Key Metrics (from PCO)
% Actively Serving	PCO	`pco_serving_teams_membership` active ÷ total congregation members
High-Frequency Servers (≥4/month)	PCO	`pco_serving_attendance` COUNT ≥ 4 in past month — burnout risk
Moderate-Frequency Servers (1-3/month)	PCO	COUNT 1-3 in past month — consistent servers
Low-Frequency Servers (0/month)	PCO	Members in teams but COUNT=0 in past month — capacity available
CN Serving data	Chinese Power BI	CN-specific serving data (supplementary)
CN Volunteer metrics	Chinese Power BI	CN-specific volunteer counts (supplementary)

Pillar 5: Celebration

> *From User Requirement Module 7: "What was this week's Celebration attendance, and how did it change percentage-wise compared to the previous week?"*

Dashboard Report	Source	Key Metrics (from PCO)
Celebration Attendance In-Person & Livestream	PCO	`pco_checkin_attendance` (in-person) + `pco_checkin_event_time` cet_total_count; livestream from separate source
EN/BM/CN/MM/NP/TM Celebration Attendance	PCO	Check-in counts by service → congregation mapping
NXG / Activities Attendance	PCO	Check-in events filtered by NXG service types
Week-over-Week % Change	PCO	Current week headcount ÷ previous week headcount
Assimilation Rate	PCO	Newcomers who completed track and joined CG ÷ total newcomers

Pillar 6: Training

> *From User Requirement Module 4: "How effectively are members engaging with and progressing through training and leadership development pathways?"*

Dashboard Report	Source	Key Metrics
Training Engagement %	PCO (pco_groups type_key=147987)	Members who attended Equip/Training class in past year ÷ total CG members
Leadership Track (L1/L2/L3) Completion	PCO (pco_groups)	Count of members who completed each leadership level
D&D Participation	PCO (pco_groups)	Members who attended specialized tracks (Church Education, Faith Formation)

Pillar 7: Mission

> *From User Requirement Module 6: "What is the level of missional training and participation in XX congregation?"*

Dashboard Report	Source	Key Metrics
MCPP1 Completion %	PCO (pco_groups / pco_attendance)	Members who attended MCPP1 ÷ total congregation
MCPP2 Completion %	PCO (pco_groups / pco_attendance)	Members who attended MCPP2 ÷ total congregation
Mission Trip Participation	PCO (pco_events / pco_attendance)	Number of members involved in external mission trips

Congregation Abbreviations

Code	Full Name
EN	English
BM	Bahasa Malaysia
CN	Chinese
MM	Myanmar

Code	Full Name
NP	Nepal
TM	Tamil
FP	Filipino
NXG	Next Generation (Youth)
CC	Campus Church

Detailed: Raw → ETL → Analytics Flow

Focus areas: All **7 pillars** — Membership, Cell Group, Evangelism, Training, Serving, Mission, Celebration.

1. RAW LAYER — PCO (Planning Center Online) Source Tables

The primary data source is **PCO** (Planning Center Online), synced daily via API into the `raw` schema. These tables are loaded **as-is** with no transformations. A secondary source (**Chinese Power BI**) provides supplementary CN-specific data.

> **Source:** "Data Dictionary for AWS PCO DB" document. Table/column names match the PCO API sync.

Core: Users (Members)

```
-- raw.pco_users
-- One row per person in the church system (members, visitors, staff, etc.)
user_id          INTEGER PK
user_first_name  VARCHAR(100) NOT NULL
user_last_name   VARCHAR(100) NOT NULL
user_name        VARCHAR(50)  NOT NULL     -- display name
user_given_name  VARCHAR(50)               -- preferred name
user_nickname    VARCHAR(50)
user_gender      VARCHAR(7)                -- NULL, 'Male', 'Female'
user_birthdate   VARCHAR(100)              -- date of birth (string format)
user_anniversary VARCHAR(100)
user_child       VARCHAR(100) NOT NULL     -- ' ' or '1' (is child)
user_membership  VARCHAR(100)              -- NULL, 'Church Member', 'Cell Member',
                                           --   'Visitor', 'Non-Attender', 'Affiliates',
                                           --   'Equipping Attender', 'Event Attender',
                                           --   'Participant', '(Archived) Church Member'
user_status      VARCHAR(100) NOT NULL     -- 'active' or 'inactive'
user_role        VARCHAR(50)               -- NULL, 'Pastor', 'Elder', 'Zone Leader',
                                           --   'Zone Pastor', 'Cell Group Leader',
                                           --   'Assistant Cell Group Leader',
                                           --   'Zone Leader | Pastor', 'Zone Leader | Elder',
                                           --   'Pastor | Elder'
user_link        VARCHAR(200) NOT NULL
user_created_at  VARCHAR(50)  NOT NULL
user_work_title  VARCHAR(500)
user_industry    VARCHAR(50)
user_baptized    VARCHAR(10)               -- NULL, 'Yes', 'Baptized', 'Baptised', 'HTBB', date
user_baptism_date VARCHAR(20)
user_medical_notes VARCHAR(1000)
```

Cell Group: Groups & Membership

```
-- raw.pco_groups
-- One row per group (Cell Groups, Leaders, Training, Ministry, etc.)
grp_id           INTEGER PK
grp_name         VARCHAR(100) NOT NULL
grp_type_key     INTEGER NOT NULL          -- 137505=Cell Groups, 162273=Leaders,
                                           --   147987=Training, 424122=Others,
                                           --   221572=Initiative, 327760=Archived,
                                           --   151990=PCO Group Training, 150510=Ministry
grp_link         VARCHAR(500) NOT NULL
grp_location     INTEGER
grp_contact_email VARCHAR(100)
grp_leader_user_key INTEGER               -- FK to pco_users (group leader)
grp_tag_zone     VARCHAR(100) NOT NULL     -- e.g. 'EN PJS1', 'CN NXG'
```

```
grp_tag_congregation VARCHAR(100)          -- 'EN - English', 'BM - Bahasa', 'CN - Chinese',
                                           --   'TM - Tamil', 'MM - Myanmar', 'NP - Nepali',
                                           --   'FP - Filipino', or ' '
grp_tag_age_group VARCHAR(100)             -- '12 and below', '13 - 17', '18 - 23',
                                           --   '24 - 35', '36 - 50', '66 and above', or ' '
grp_tag_freq     VARCHAR(100)              -- 'Weekly', 'Monthly', 'Twice Monthly', 'Varied', ' '
grp_tag_ministry VARCHAR(100)              -- e.g. 'CN Facility', 'EQ Level 2'
grp_size         INTEGER NOT NULL
grp_status       VARCHAR(20)               -- NULL, 'NEW', 'ARCHIEVED'
grp_zone_leader  VARCHAR(200)              -- e.g. 'EN MD - ZL: Pr Gilbert'
grp_tag_district VARCHAR(100)              -- e.g. 'BM - District 1'
grp_tag_subzone  VARCHAR(100) NOT NULL     -- e.g. 'CN ZL: Pr Samuel Tan'
grp_location_type_preference VARCHAR(20)   -- NULL

-- raw.pco_group_membership
-- Junction: which users belong to which groups
member_id        INTEGER PK
member_joined_date TIMESTAMPTZ NOT NULL
member_role      VARCHAR(20) NOT NULL      -- 'member' or 'leader'
member_grp_key   INTEGER FK (pco_groups)
member_user_key  INTEGER FK (pco_users)
member_user_link VARCHAR(500) NOT NULL
member_grp_link  VARCHAR(500) NOT NULL
```

Cell Group & Celebration: Events & Attendance

```
-- raw.pco_events
-- One row per scheduled event (CG meetings, celebrations, activities)
event_id         INTEGER PK
event_name       VARCHAR(200) NOT NULL
event_group_key  INTEGER FK (pco_groups)   -- which group this event belongs to
event_location_key INTEGER
event_startdatetime VARCHAR(30) NOT NULL
event_enddatetime VARCHAR(100) NOT NULL
event_canceled   VARCHAR(20) NOT NULL      -- '0' (not cancelled) or '1' (cancelled)
event_canceled_at VARCHAR(100)
event_reminders_sent VARCHAR(20) NOT NULL  -- '0' or '1'
event_reminders_sent_at VARCHAR(20)
event_description VARCHAR(4000)
event_att_committed_at VARCHAR(30)         -- when attendance was submitted (NULL = not submitted)
event_job_key    INTEGER
event_location_type_preference VARCHAR(20)

-- raw.pco_attendance
-- One row per user per event attended (CG + celebration attendance)
att_event_key    INTEGER FK (pco_events)
att_group_key    INTEGER FK (pco_groups)
att_user_key     INTEGER FK (pco_users)
att_user_role    VARCHAR(20) NOT NULL      -- 'member', 'visitor', 'leader', 'applicant'
att_status       VARCHAR(20) NOT NULL      -- '0' (didn't attend) or '1' (attended)
att_job_date     VARCHAR(50)               -- date data was pulled from PCO
```

Celebration: Check-in System

```
-- raw.pco_checkin_events
-- Check-in event definitions (celebration services, kids church, etc.)
ce_id            INTEGER PK
ce_name          VARCHAR(1000) NOT NULL
ce_frequency     VARCHAR(50) NOT NULL      -- 'Daily', 'Weekly', 'None'
ce_archived_at   VARCHAR(200)

-- raw.pco_checkin_event_time
-- Timing details for check-in events
cet_id           INTEGER PK
cet_day_of_week  INTEGER NOT NULL          -- 0=Sunday, 1=Monday, ..., 6=Saturday
cet_guest_count  INTEGER NOT NULL
cet_hour         INTEGER NOT NULL          -- 24H format
cet_minute       INTEGER NOT NULL
cet_shows_at     TIMESTAMP NOT NULL
```

```
cet_start_at     TIMESTAMPTZ NOT NULL
cet_total_count  INTEGER NOT NULL          -- headcount
cet_ce_event_key INTEGER FK (pco_checkin_events)
cet_eventperiod_key INTEGER NOT NULL

-- raw.pco_checkin_attendance
-- Individual check-in records (in-person celebration attendance)
ca_id            INTEGER PK
ca_checked_out_at TIMESTAMP
ca_confirmed_at  TIMESTAMP
ca_first_name    VARCHAR(200) NOT NULL
ca_last_name     VARCHAR(200) NOT NULL
ca_kind          VARCHAR(50) NOT NULL      -- 'Guest', 'Regular', 'Volunteer'
ca_number        INTEGER NOT NULL
ca_user_key      INTEGER FK (pco_users)
ca_eventperiod_key INTEGER NOT NULL
ca_ce_key        INTEGER FK (pco_checkin_events)
ca_checked_in_by_user_key INTEGER
ca_checked_out_by_user_key INTEGER
ca_created_at    TIMESTAMPTZ NOT NULL
```

Evangelism: Decisions & Discipleship

```
-- raw.pco_new_decisions
-- Evangelism decision form submissions
nd_id            INTEGER PK
nd_user_key      INTEGER FK (pco_users)
nd_user_name     VARCHAR(255) NOT NULL
nd_form_submitted INTEGER NOT NULL         -- 112029 = 'EN - Decision Form' (outside church)
                                           -- 117917 = 'EN - Decision Form (Online)' (during celebration)
nd_form_title    VARCHAR(255) NOT NULL
nd_submitted_at  DATE NOT NULL
nd_reason        VARCHAR(1000) NOT NULL    -- concatenated answers from the form

-- raw.pco_discipleship_completion
-- Discipleship workflow tracking (D101, new believer follow-up)
dc_id            INTEGER PK
dc_user_key      INTEGER FK (pco_users)
dc_user_name     VARCHAR(255) NOT NULL
dc_started_at    DATE NOT NULL             -- date added to workflow
dc_workflow_completion VARCHAR(50) NOT NULL -- 'In progress' or 'Completed'
dc_completed_at  DATE                      -- NULL if not yet completed
```

Serving: Teams & Attendance

```
-- raw.pco_serving_teams
-- Serving team definitions
st_team_id       INTEGER PK
st_team_name     VARCHAR(255) NOT NULL     -- e.g. 'Worship Team - Musicians'
st_service_type  VARCHAR(255) NOT NULL     -- parent ministry, e.g. 'DUMC EN NextGen'

-- raw.pco_serving_teams_membership
-- Which users belong to which serving teams
stm_id           INTEGER PK
stm_user_key     INTEGER FK (pco_users)
stm_team_key     INTEGER FK (pco_serving_teams)
stm_user_name    VARCHAR(255) NOT NULL

-- raw.pco_serving_attendance
-- When volunteers actually served
sa_id            INTEGER PK
sa_plan          INTEGER NOT NULL          -- unique event/plan ID
sa_user_key      INTEGER FK (pco_users)
sa_team_key      INTEGER FK (pco_serving_teams)
sa_user_name     VARCHAR(255) NOT NULL
sa_team_name     VARCHAR(255) NOT NULL
sa_service_type  VARCHAR(255) NOT NULL     -- parent ministry
```

```
sa_position      VARCHAR(255) NOT NULL     -- specific role (e.g. 'Drums')
sa_service_date  DATE NOT NULL
```

Infrastructure: Room Booking & ETL Config

```
-- raw.pco_rooms / raw.pco_room_booking
-- Room management (not used in semantic layer, retained for reference)

-- raw.run_config / raw.run_log
-- ETL job scheduling and execution logs
-- run_config: defines sync jobs (daily/weekly), curl commands, insert SQL
-- run_log: tracks job execution status ('RUNNING', 'DONE')
```

2. ETL LAYER — Cron Job + SQL Scripts

Each transformation is a **SQL script** executed by a cron job in dependency order: **dimensions first, then facts**.

> **Note:** Two source systems feed into the raw layer:
>
> - **PCO (Planning Center Online)** — primary system for all congregations, synced via API into `pco_*` tables
> - **Chinese Power BI** — supplementary CN-specific data (CG metrics, serving, volunteers)
>
> The ETL must handle deduplication where CN data appears in both sources.

> **Note:** If ETL complexity grows (e.g., more fact tables, complex dependencies, need for retry/alerting), consider migrating to Alibaba Cloud DataWorks for built-in scheduling, dependency management, and monitoring.

```
Cron Job Execution Order (daily schedule):

DIMENSIONS (run in parallel — no dependencies between them):
┌──────────┐  ┌──────────┐  ┌──────────┐
│ dim_date │  │dim_member│  │  dim_cg  │
└──────────┘  └──────────┘  └──────────┘
     │            │            │
     └────────────┴────────────┘
                  │
                  ▼
  dim_congregation = seed data (no ETL)

FACTS (run after dimensions):
┌──────────────────┐  ┌──────────────────┐  ┌──────────────────┐
│fact_cg_attendance│  │  fact_decisions  │  │fact_celebration  │
└──────────────────┘  └──────────────────┘  │  _attendance     │
         │                    │             └──────────────────┘
         │                    │                       │
┌────────▼─────────┐  ┌───────▼──────────┐  ┌────────▼─────────┐
│fact_cg_submission│  │fact_discipleship │  │  fact_serving    │
└──────────────────┘  └──────────────────┘  └──────────────────┘
```

ETL Step	Script File	PCO Source Tables	Description	Refresh
1. dim_date	`03-etl-transforms/01-dim-date.sql`	*(generated)*	Date spine: year, quarter, month, week_of_year, day_of_week, weekend_flag	INSERT IGNORE (append only)
2. dim_member	`03-etl-transforms/02-dim-member.sql`	`pco_users` + `pco_group_membership` + `pco_groups`	Parse birthdate→age_group, map user_role, user_membership, derive congregation_key from group tags, is_baptized from user_baptized	Full refresh (truncate + reload)
3. dim_cell_group	`03-etl-transforms/03-dim-cell-group.sql`	`pco_groups` (grp_type_key=137505) + `pco_group_membership`	CGs with zone, district, subzone, zone_leader (all denormalized from group tags), leader_member_key, congregation_key, age_group, frequency, size, status	Full refresh
4. dim_congregation	*(no ETL — seed data)*	—	8-row reference table (EN/BM/CN/MM/NP/TM/FP/NXG).	Static seed

ETL Step	Script File	PCO Source Tables	Description	Refresh
			Seeded in CREATE TABLE script, not refreshed.	
5. fact_cg_attendance	`03-etl-transforms/04-fact-cg-attendance.sql`	`pco_attendance` + `pco_events` + `pco_groups`	Filter: grp_type_key=137505 (CG events only). Map att_status→attended, att_user_role→user_role, event_startdatetime→date_key	Incremental (watermark on att_job_date)
6. fact_cg_submission	`03-etl-transforms/05-fact-cg-submission.sql`	`pco_events` + `pco_groups`	Filter: grp_type_key=137505. event_att_committed_at IS NOT NULL→submitted, event_canceled='1'→cancelled	Incremental (watermark on event_startdatetime)
7. fact_decisions	`03-etl-transforms/06-fact-decisions.sql`	`pco_new_decisions` + `pco_users`	Map nd_submitted_at→date_key, nd_form_submitted→form_id, derive congregation_key from user's group	Incremental (watermark on nd_submitted_at)
8. fact_discipleship	`03-etl-transforms/07-fact-discipleship.sql`	`pco_discipleship_completion`	Map dc_started_at→started_date_key, dc_completed_at→completed_date_key, dc_workflow_completion→status	Full refresh
9. fact_celebration_attendance	`03-etl-transforms/08-fact-celebration-attendance.sql`	`pco_checkin_attendance` + `pco_checkin_events`	Map ca_created_at→date_key, ca_kind→kind, ce_name→service_name, derive congregation_key from service name	Incremental (watermark on ca_created_at)
10. fact_serving	`03-etl-transforms/09-fact-serving.sql`	`pco_serving_attendance` + `pco_serving_teams`	Map sa_service_date→date_key, sa_user_key→member_key, join team name/ministry/position	Incremental (watermark on sa_service_date)

Cron runner script: `05-dataworks-config/run-etl.sh`

3. SEMANTIC LAYER — Star Schema Tables

These are the target tables the BI tool reads from. Full CREATE TABLE definitions in:

- `02-semantic-layer/01-create-sem-schema.sql` — sem schema (4 dimensions + 6 facts)

Table	Type	Grain	Pillars Served
`dim_date`	Dimension	One row per calendar date (year, quarter, month, week_of_year, day_of_week, weekend_flag)	All
`dim_member`	Dimension	One row per member (age_group, gender, user_role, membership, congregation_key, is_baptized)	All
`dim_cell_group`	Dimension	One row per CG (zone, district, subzone, zone_leader, leader, congregation_key, age_group, frequency, size, status)	Cell Group, Membership
`dim_congregation`	Dimension	8-row seed table (EN/BM/CN/MM/NP/TM/FP/NXG → labels)	All
`fact_cg_attendance`	Fact	One row per member per CG event (attended, user_role)	Cell Group
`fact_cg_submission`	Fact	One row per CG per event (submitted, cancelled)	Cell Group
`fact_decisions`	Fact	One row per decision form (form_id, form_title, congregation_key)	Evangelism
`fact_discipleship`	Fact	One row per participant (status, started_date_key, completed_date_key)	Evangelism, Training
`fact_celebration_attendance`	Fact	One row per check-in (kind, service_name, congregation_key)	Celebration
`fact_serving`	Fact	One row per volunteer per serve date (team_name, ministry, position)	Serving

Data Lineage Summary

```
PCO RAW TABLE                    →  ETL TRANSFORM                  →  ANALYTICS TABLE
```

```
DIMENSIONS:
(generated)                        →  date spine generation            →  dim_date
pco_users                          →  parse birthdate→age_group,       →  dim_member
  + pco_group_membership           →  map user_role, user_membership,  →    (age_group, gender, role,
  + pco_groups                     →  derive congregation from groups  →     membership, congregation_key,
                                   →  parse user_baptized→is_baptized  →     is_baptized)
pco_groups (type_key=137505)       →  parse tags: zone, district,      →  dim_cell_group
  + pco_group_membership           →  subzone, zone_leader, congr.,    →    (zone, district, subzone,
                                   →  age_group, freq, size, status    →     zone_leader, leader, size)
(seed data — no ETL)               →  static 8-row insert              →  dim_congregation

FACTS:
pco_attendance                     →  filter: grp_type_key=137505      →  fact_cg_attendance
  + pco_events                     →  att_status→attended              →    (member_key, cg_key, date_key,
  + pco_groups                     →  att_user_role→user_role          →     attended, user_role)
pco_events                         →  filter: grp_type_key=137505      →  fact_cg_submission
  + pco_groups                     →  event_att_committed_at→submitted →    (cg_key, date_key,
                                   →  event_canceled→cancelled         →     submitted, cancelled)
pco_new_decisions                  →  nd_form_submitted→form_id        →  fact_decisions
  + pco_users                      →  derive congregation_key          →    (member_key, date_key,
                                   →                                   →     form_id, congregation_key)
pco_discipleship_completion        →  dc_started_at→started_date_key   →  fact_discipleship
                                   →  dc_completed_at→completed_date_key   (member_key, status,
                                   →  dc_workflow_completion→status    →     started, completed)
pco_checkin_attendance             →  ca_kind→kind                     →  fact_celebration_attendance
  + pco_checkin_events             →  ce_name→service_name             →    (member_key, date_key, kind,
                                   →  derive congregation_key          →     service_name, congregation_key)
pco_serving_attendance             →  sa_service_date→date_key         →  fact_serving
  + pco_serving_teams              →  join team name, ministry, pos.   →    (member_key, date_key, team,
                                   →                                   →     ministry, position)
```

Dashboard Queries by Pillar

See full query files organized by pillar:

Pillar 1: Membership

- `04-dashboard-queries/01-membership/01-cg-size-by-year.sql` — DUMC CG Size by Year (from dim_cell_group)
- `04-dashboard-queries/01-membership/02-staff-demographics.sql` — Staff demographics (dim_member WHERE user_role LIKE '%Pastor%' or staff)
- `04-dashboard-queries/01-membership/03-pastor-demographics.sql` — Pastor demographics (dim_member WHERE user_role IN ('Pastor', 'Zone Pastor'...))
- `04-dashboard-queries/01-membership/04-cg-leader-demographics.sql` — CG Leader demographics (dim_member WHERE user_role = 'Cell Group Leader')
- `04-dashboard-queries/01-membership/05-cg-leader-age-gender.sql` — CG Leader age/gender (age_group × gender pivot)
- `04-dashboard-queries/01-membership/06-membership-ratio.sql` — Church Member % / CG Member % / Lost Sheep %

Pillar 2: Cell Group

- `04-dashboard-queries/02-cell-group/01-weekly-cg-attendance-all.sql` — Weekly CG Attendance [All Congregation]
- `04-dashboard-queries/02-cell-group/02-cg-health-analysis.sql` — CG Health Analysis by congregation (EN/BM/CN/MM/NP/TM)
- `04-dashboard-queries/02-cell-group/03-weekly-cg-attendance-by-zone.sql` — Weekly CG Attendance Analysis By Zone
- `04-dashboard-queries/02-cell-group/04-avg-cg-attendance-by-zone.sql` — Average CG Attendance By Zone
- `04-dashboard-queries/02-cell-group/05-weekly-cg-attendance-rate.sql` — Weekly CG Attendance Rate By CG
- `04-dashboard-queries/02-cell-group/06-cg-submission-rate.sql` — CG Submission Rate
- `04-dashboard-queries/02-cell-group/07-non-submission-cg.sql` — Non-submission of CG Attendance
- `04-dashboard-queries/02-cell-group/08-pastor-attendance-analysis.sql` — Pastor Attendance Analysis
- `04-dashboard-queries/02-cell-group/09-zone-leader-cg-attendance.sql` — Zone Leader CG Attendance Analysis
- `04-dashboard-queries/02-cell-group/10-active-cg-sessions.sql` — Weekly Count of Active CG Sessions
- `04-dashboard-queries/02-cell-group/11-cn-cg-metrics.sql` — CN Cell Group metrics (Chinese Power BI)
- `04-dashboard-queries/02-cell-group/12-cn-cg-attendance.sql` — CN CG Attendance (Chinese Power BI)

Pillar 3: Evangelism

- `04-dashboard-queries/03-evangelism/01-decisions-churchwide.sql` — DUMC Decisions Churchwide (COUNT from fact_decisions)
- `04-dashboard-queries/03-evangelism/02-decisions-by-congregation.sql` — Decisions by congregation (BM/CN/EN/MM/NP/TM/FP)
- `04-dashboard-queries/03-evangelism/03-en-nxg-cc-decisions.sql` — EN NXG CC Decisions
- `04-dashboard-queries/03-evangelism/04-baptisms-by-congregation.sql` — Baptisms (dim_member WHERE is_baptized=1)

- `04-dashboard-queries/03-evangelism/05-d101-completions.sql` — D101 Completions (fact_discipleship WHERE status='Completed')
- `04-dashboard-queries/03-evangelism/06-decision-form-breakdown.sql` — Outside (112029) vs Online (117917) form breakdown

Pillar 4: Serving

- `04-dashboard-queries/04-serving/01-actively-serving-pct.sql` — % Actively Serving by congregation
- `04-dashboard-queries/04-serving/02-serving-frequency-tiers.sql` — High/Moderate/Low frequency server breakdown
- `04-dashboard-queries/04-serving/03-cn-serving-data.sql` — CN Serving data (Chinese Power BI supplement)
- `04-dashboard-queries/04-serving/04-cn-volunteer-metrics.sql` — CN Volunteer metrics (Chinese Power BI supplement)

Pillar 5: Celebration

- `04-dashboard-queries/05-celebration/01-celebration-attendance.sql` — Celebration Attendance by kind (Regular/Guest/Volunteer)
- `04-dashboard-queries/05-celebration/02-celebration-by-congregation.sql` — Celebration Attendance by congregation (EN/BM/CN/MM/NP/TM)
- `04-dashboard-queries/05-celebration/03-nxg-activities-attendance.sql` — NXG / Activities Attendance
- `04-dashboard-queries/05-celebration/04-week-over-week-change.sql` — Week-over-Week % change
- `04-dashboard-queries/05-celebration/05-assimilation-rate.sql` — Newcomer assimilation rate (Guest→CG member)

Pillar 6: Training

- `04-dashboard-queries/06-training/01-training-engagement.sql` — Training Engagement % (attended Equip/Training in past year ÷ CG members)
- `04-dashboard-queries/06-training/02-leadership-track-completion.sql` — Leadership Track L1/L2/L3 completion counts
- `04-dashboard-queries/06-training/03-dnd-participation.sql` — D&D / specialized track participation

Pillar 7: Mission

- `04-dashboard-queries/07-mission/01-mcpp1-completion.sql` — MCPP1 Completion % by congregation
- `04-dashboard-queries/07-mission/02-mcpp2-completion.sql` — MCPP2 Completion % by congregation
- `04-dashboard-queries/07-mission/03-mission-trip-participation.sql` — Mission Trip Participation count

Validation

- `04-dashboard-queries/99-validation/01-row-count-checks.sql` — row count checks across layers
- `04-dashboard-queries/99-validation/02-source-reconciliation.sql` — NCG vs Chinese Power BI dedup checks
- `04-dashboard-queries/99-validation/03-orphan-detection.sql` — orphan/missing key detection

Steps to Implement

Step 1: Finalize data migration to RDS

- Complete the migration of PCO tables into Alibaba Cloud RDS (see "Data Dictionary for AWS PCO DB")
- Key PCO tables: `pco_users`, `pco_groups`, `pco_group_membership`, `pco_events`, `pco_attendance`, `pco_checkin_attendance`, `pco_checkin_events`, `pco_new_decisions`, `pco_discipleship_completion`, `pco_serving_attendance`, `pco_serving_teams`, `pco_serving_teams_membership`
- Set up daily API sync using existing `run_config` / `run_log` infrastructure
- Integrate Chinese Power BI CN-specific data as supplementary tables

Step 2: Design the star schema (data model)

- Review existing PCO data + Chinese Power BI dashboards to confirm all metrics are captured
- Map out fact tables and dimension tables as detailed in the "Raw → ETL → Analytics Flow" section above
- Document the grain of each fact table (see "Our Six Fact Tables" section)
- Start with the **Cell Group** pillar (highest number of metrics) then expand to other pillars
- Ensure CN data from Chinese Power BI integrates cleanly with NCG data (dedup strategy)

Step 3: Set up cron job for ETL

- Create a shell script that runs the ETL SQL scripts in order (dimensions first, then facts)
- Schedule via cron on an ECS instance (e.g., daily at 4:00 AM before staff arrive)
- Add logging and basic email alerts on failure
- Consider upgrading to Alibaba Cloud DataWorks later if scheduling/monitoring needs grow

Step 4: Create the semantic layer

- Create a `sem` schema in RDS
- Deploy the star schema tables with proper indexes

- Run the ETL cron job to populate them
- Validate data against existing dashboards for accuracy

Step 5: Connect BI tool and rebuild dashboards

- Connect Power BI / QuickBI / new vendor to the `sem` schema (semantic layer)
- Recreate existing dashboard reports pointing to the star schema
- Rebuild dashboards organized by the **7 Pillars**:
  - **Membership**: CG size trends, staff/pastor/CG leader demographics
  - **Cell Group**: Weekly attendance (all/by zone/by CG), health analysis, submission rates, pastor & zone leader attendance, active session counts, CN-specific metrics
  - **Evangelism**: Churchwide decisions, decisions by congregation, NXG CC decisions
  - **Serving**: % actively serving, frequency tiers, CN serving/volunteer data
  - **Celebration**: Attendance by kind (Regular/Guest/Volunteer), by congregation, WoW change, assimilation
  - **Training**: Training engagement %, leadership track L1/L2/L3, D&D participation
  - **Mission**: MCPP1/MCPP2 completion %, mission trip participation
- Validate that numbers match the old dashboards

Step 6: Testing and go-live

- Run old and new dashboards in parallel for 1-2 weeks
- Compare outputs to catch discrepancies
- Cut over once confident

Cost Estimate

Service	Estimated Cost
RDS (already have)	Existing cost
Cron job (on existing ECS)	$0 (runs on existing infra)
DataWorks (if needed later)	Free tier likely sufficient; paid starts ~$50/mo
QuickBI (if chosen)	Starts ~$30/user/mo
Total additional	**~$0/mo** (cron) or **~$0-50/mo** (with DataWorks)

Using a separate RDS schema instead of AnalyticDB keeps costs minimal — ideal for a church budget.

Verification

- After Step 4: Run sample queries against sem tables and compare with raw data
- After Step 5: Side-by-side comparison of old vs new dashboards
- Ongoing: Monitor cron job logs for daily ETL success/failure